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ORIGINAL ARTICLE

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Environmental variables for modeling wheat yields in the southwest pampa region of Argentina

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Abstract Two types of time scales – 10-day intervals (D) and phenological phases (P) – were applied to environmental variables for the development of statistical regression models relating to the southwest pampa region of Argentina, with the aim of detecting the effects of weather on wheat yields for the period 1977–1999. The parameters were grouped as meteorological and processed variables and indices. The processed variables used were total soil water availability (SWA) and the ratio of actual evapotranspiration to potential evapotranspiration (α), obtained from a water-balance model in which the moisture anomaly index (Z) and the Palmer drought severity index (PDSI) were calculated according to Palmer's model. For these parameters it was possible to establish the times of year and the phenological phases with the best correlation to grain yield. The regression equation for meteorological variables on a 10-day scale provides one of the best fits. Using mixed parameters, the two models, D and P, give rise to a standard error of estimate of approximately 200 kg ha^{-1}. Truncated models perform better on a P scale than on a D scale. The use of phenological stages improved yield assessment, particularly for those years with extreme meteorological conditions. The optimum models were tested and root-mean-square errors (RMSE) of 440 kg ha^{-1} and 470 kg ha^{-1} were obtained for P and D scales respectively.

Keywords Wheat modeling · Multivariate regression · Environmental variables · Pampa region

Introduction

The southwest border of the pampa region is characterized by a subhumid climate, high interannual rain variability and a negative soil moisture balance. The study area is located on a thermic, loamy sand, Entic Haplustoll soil. The effective depth varies between 0.8 m and 1.8 m, determined by the petrocalcic horizon. A climatic aridity index as defined in Hatfield (1990) was selected to identify the area. The semiarid zone corresponds to index values below 0.5 whereas the subhumid regions lie between 0.5 and 0.75. Figure 1 shows part of the pampa region, the aridity index and the location of study sites.

Wheat is the main cereal grown in the region, and variable weather conditions have a direct impact on crop yield. The principle environmental factors affecting wheat development are temperature, photoperiod and vernalization (Miralles and Slafer 2001). Other factors that influ-

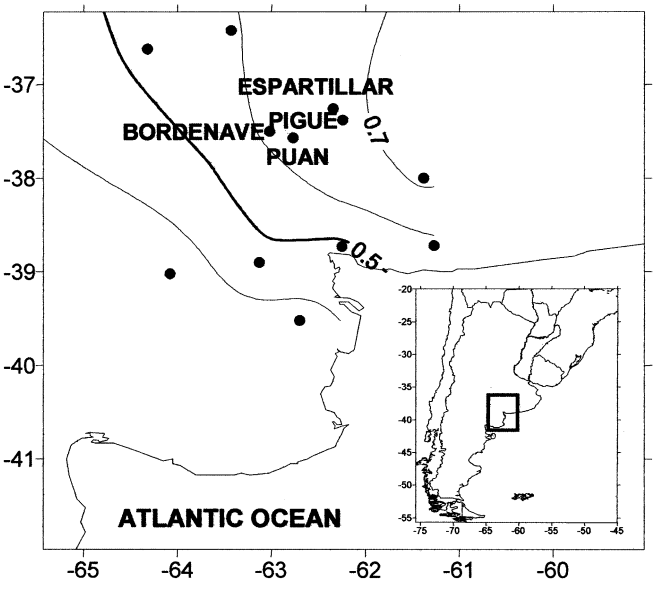

Fig. 1 Location of the studied pampa region, climatic aridity index (–) and cited study sites together with non-labeled locations (•) employed to compute the aridity index. (Negative latitude and longitude values correspond to positions South and West respectively)

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ence growth and productivity are water availability and radiation. Several authors have pointed out that winter cereals in the region suffer from water stress during the late spring season. The highest correlation between monthly precipitation and yearly wheat yield is observed in November, March and April (Paoloni and Vazquez 1984). Other authors recognize the yield-reducing effect of low initial soil moisture content, high air temperature in the filling stage and water deficiency during anthesis (Travasso et al. 1994). Average yields for the southwest pampa region are 2,000 kg ha^{-1}, with maximum values reaching 3,500–3,600 kg ha^{-1} in cases of early and abundant rainfall, no late frosts, and moderate temperatures during grain filling (Gallo Candolo 2001).

Regression-type models have been used to assess weather effects on yield since the beginning of the 20th century, though they have been criticized for their empirical nature, which restricts their application to the specific range of values for which they were developed. Nevertheless they are still useful in regions where it is not possible to use simulation or deterministic growth models owing to the inability to determine the model parameters correctly. Three crop-simulation models recently used to test the accuracy of grain yield predictions under UK climatic conditions failed to successfully predict wheat yield. The inaccurate results may have been due in part to inadequately specified parameters and errors in weather variables or sowing dates (Landau et al. 1998). The same authors were able to improve their results when they developed a parsimonious, multiple-regression model, using a minimum number of parameters without losing predictive power (Landau et al. 2000). There is also renewed interest in this type of model as a tool for assessing the impact of climatic variations on crop production over widespread areas. Research aimed at assessing wheat yields was carried out in Bordenave by Scian and Donnari (1995), using the monthly Palmer drought severity index (PDSI) and the moisture anomaly index (Z).

In the present study we used weather data from an experimental site in the southwest of the pampa region to study the suitability of a multivariate model of wheat yield prediction, employing 10-day intervals and phenological stages.

Data and methods

Wheat yield and meteorological data from Bordenave (latitude: 37°51′S, longitude: 63°01′W, elevation: 212 m), Province of Buenos Aires, were used (see Fig. 1). Estimated grain yields for other locations were also used. Palmer's model (Palmer 1965) was applied on a 10-day time scale, from 1977 to 2000. The parameters used were accumulated rainfall (P_p), minimum and maximum mean temperature (T_n and T_x), total soil water storage (SWA), the ratio between potential and actual evapotranspiration (PET/AET or α) as obtained from Palmer's water-balance model, and the moisture anomaly (Z) and Palmer drought severity (PDSI) indices. Classical Pearson correlation formulae were applied with a 5% significance level.

A multiple-regression model of the type

$$Y = \beta_0 + \sum \beta_i X_i, \quad i = 1, \ldots k \tag{1}$$

where Y denotes observed yield and X_i the respective environmental variable, was used as the basic model (Draper and Smith 1981). Different cases were analyzed using meteorological variables, processed variables, indices, and combinations of these. Variables such as management, genetics, and technology were not included. The total model represents the selection of the optimum variables and the truncated model is generated when a restriction is imposed on the number of 10-day intervals included. The time period spans from the pre-sowing stage (April and May) to harvest time (December). Variables were pre-selected using a forward stepwise method followed by a backward mode. Finally five to six optimum variables were selected in accordance with the determination coefficient (R^2) and the standard error of estimate (SEE). Care was taken to avoid variable redundancy. The first 16 years of crop yields were used to obtain the models and the rest to test them, using the root-mean-square error (RMSE) because of it is more sensitive to larger error than is the mean absolute error (MAE).

Wheat yields from the experimental field correspond to full-season wheat cultivars, annual cropping and a fixed sowing date. Taking into account the growth patterns of wheat and suitable periods for 10-day agroclimatic data, the following phenological stages and 10-days intervals were used:

1. Sowing: from 21 June to 10 July (jn3–jl1); most planting in the region is done during this period.
2. Emergence: from 11 July to 30 August (jl2–ag3).
3. Tillering: from 1 September to 30 September (se1–se3).
4. Stem elongation: from 1 October to 20 October (oc1–oc2).
5. Heading: the last 10 days of October (oc3).
6. Anthesis or flowering: the first 10 days of November (no1).
7. Grain filling: from 11 November to 30 November (no2–no3).
8. Maturity: the two first 10-day periods of December (de1–de2).
9. Harvest: the last 10-days of December (de3).

The abbreviations used to define the 10-day period consist of three characters: the two letters represent the month and the number (1, 2 or 3) indicates the first, second or third 10-day interval of the month (as in Tables 1–4 and Fig. 2).

Results

Data analysis

Observed wheat yields in Bordenave and estimated yields from three neighboring locations less than 50 km from Bordenave (Espartillar, Puan, and Pigue) were analyzed on a year-by-year basis (see Fig. 2). The average yield for the period 1979–2000 amounted to 1,600 kg ha^{-1}. Two mean maxima can be distinguished, one for 1984/1985 (2,500 kg ha^{-1}) and one for 1997/1998 (2,560 kg ha^{-1}).

The correlation coefficients between wheat yields in Bordenave and the other sites are highly significant, with maximum and minimum values ranging from 0.76 to 0.89. It therefore appears possible to extend the application of the models based on regression equations using Bordenave data to the region as a whole, thus explaining 60%–80% of regional yield variability.

In order to characterize the study area, mean values of the meteorological variables (T_x, T_n and P_p) at Bordenave for 10-day intervals from 1971 to 2000 are shown in Fig. 3 together with the wheat cycle (June to December).

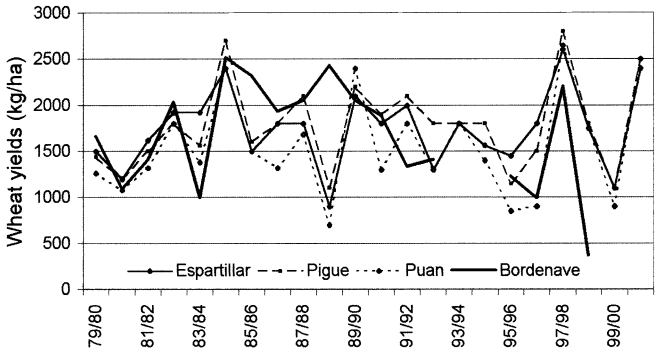

Fig. 2 Estimated (for three locations in the southwest pampa) and observed (Bordenave) wheat yields (kg ha^{-1}) during 1979–2000 crop years

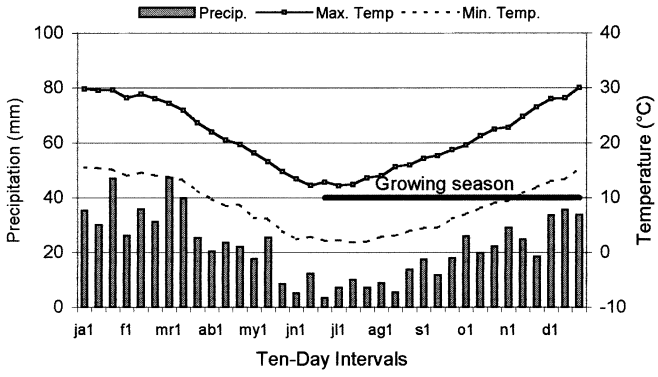

Fig. 3 Ten-day interval means for accumulated precipitation and mean maximum and minimum temperatures at Bordenave (1971–2000). The growing season for wheat in the region ranges from late June to December

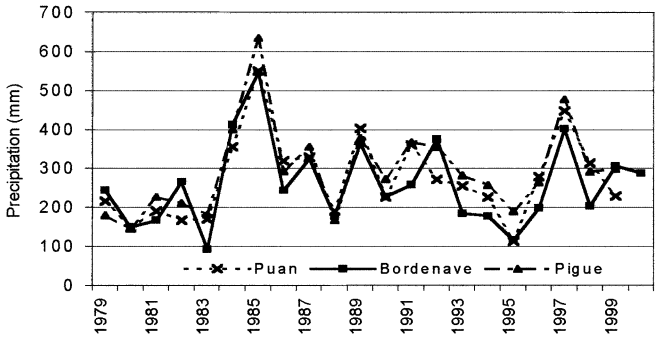

Fig. 4 Accumulated precipitation (mm) during the crop season at Bordenave. The same for two other locations in Buenos Aires Province. Period 1979–2000

The annual mean and standard deviation for rainfall are 790 ± 183 mm. Ten-day mean maximum temperatures do not exceed 30 °C and minimum mean values are positive. During crop development both temperature and rainfall increase.

The incidence of rainfall during the wheat season is given in Fig. 4. Maximum accumulated precipitation for the crop cycle occurred in 1984–1985 and 1997, coinciding with maximum yield. Increased rainfall during the 1970s resulted in higher yields (Sierra and Brynsztein 1990; Castañeda and Barros 1994; Roberto et al. 1994; Hoffmann et al. 1997). During the study period described here, however, there was no clear positive rainfall trend,

Table 1 Agroclimatic 10-day variables best related to wheat yields at Bordenave for the full model (×) and the pre-season model (pre-season and crop-cycle months, shown by shading). Meteorological parameters (i): P_p precipitation, T_n minimum temperature, T_x maximum temperature. Processed parameters (ii): *SWA* total soil water storage, α potential to actual evapotranspiration. Index parameters (iii): *Z* moisture anomaly index, *PDSI* Palmer drought severity index. Enhanced 10-day intervals (*bold type*) indicate the largest number of simultaneously significant parameters. Coefficients of determination for the two regression models and for each parameter are also shown (Roman numerals indicate parameters in decreasing order of R^2 values)

Interval	P_p(i)	Tmi(i)	Tx(i)	SWA(ii)	α (ii)	Z(iii)	PDSI (iii)
ap1						X	X
ap2							
ap3						X	
my1				X			
my2						X	
my3			X				
jn1				X	X		
jn2							
jn3		X					
jl1							
jl2							
jl3		X					
ag1	X		X	X	X		X
ag2						X	X
ag3							X
se1		X					
se2		X					
se3		X					
oc1	X		X	X	X		
oc2	X		X	X	X	X	
oc3			X		X		
no1	X		X	X		X	X
no2	X			X			
no3							X
de1							
de2							
de3							
Adj R^2	0.8211	0.8932	0.6643	0.9066	0.9696	0.9463	0.7498
R^2	0.8964	0.9497	0.7566	0.9417	0.9826	0.9718	0.8420
Full model	V	III	VII	IV	I	II	VI
Adj R^2	0.7220	0.7424	0.7648	0.8498	0.9138	0.9235	0.8602
R^2	0.8538	0.8485	0.8893	0.9028	0.9594	0.9640	0.9178
Pre-season model	VI	VII	V	IV	II	I	III

so it must be assumed that yield variability in this period is mainly a result of natural variations in the climate.

Ten-day interval models

For the multivariate regression analysis the stepwise method was used, with an F to-enter (Fisher test value) between 1 and 3.5, except for the truncated models, and a minimum tolerance between 0.01 and 0.20, depending on the restrictions set on the selection of the variables to-enter. The variables included are meteorological parameters (P_p, T_x and T_n), surface water-balance variables (SWA and α) and indices (Z and PDSI). Two cases were analyzed. In the first trial, variables for the whole year were considered (full model), and then a pre-season period of more than 2 months was selected prior to the sowing (pre-season model). Table 1 presents the selected variables for both exploratory regression models starting on 1 April. The best-related variables are indicated (by a cross for the full model or by shading for the pre-season model). In accordance with the results for the seven linear regression variables, the ranking of the parameters in decreasing order of R^2 is α, Z, T_n, SWA, P_p, PDSI, and T_x for the full model; and Z, α, PDSI, SWA, T_x, P_p, and T_n for the pre-season model. The conclusion drawn therefore is that α and Z are the best two parameters on which to base a statistical model for the region. The first 10 days in August, the second 10 days in October and the first 10 days in November during the crop cycle correspond to the largest number of simultaneously significant variables. The statistical analysis shows that crop development during these particular periods is highly affected by thermal and hydric anomalies.

From several trials using meteorological variables, the best-fit equation includes T_n(jl2), T_n(ag3), T_n(no3), T_x(oc2), P_p(ag2) and P_p(no2) (model 1D). Minimum temperatures during August and November and maximum temperatures in October negatively affect yield. The R^2 coefficient is 0.98 and the SEE is 90.5 kg ha^{-1}.

When processed parameters are used in a regression model, the selected variables are α(oc1), α(no2), α(se2), α(ag2), SWA(oc1), α(no1) (model 2D). The value of the coefficient R^2 is 0.92, with SEE equal to 186.8 kg/ha. Taking into account the variables included in the model up to 20 November, this selection should define a naturally truncated predictor model. The inclusion of the variable SWA(oc1) shows the importance of soil moisture during stem elongation. Variable α denotes the relationship between water availability and plant requirements and is important throughout almost the entire crop cycle. According to the RMSE obtained in the predictability test, this selection does not make a good predictor model.

The index variable integrates the effect of meteorological and soil water availability parameters into one single value. The selected variables for such a model include PDSI during planting (model 3D). Other variables are: Z(se3), Z(no2), Z(ag2), Z(no3) and Z(ag1). This model has a SEE of 318 kg/ha. (See Table 2 for a summary of the results of models).

All-variables 10-day model

The forward stepwise method was used in order to undertake a general selection of variables. In accordance with the results obtained in the above paragraph, ten variables were included in the total model: α(oc1), Z(no2), Z(no3), Z(oc2), T_x(ag2), Z(no1), P_p(se3), P_p(jl2), P_p(jl3), T_n(se3). Thus, using the backward mode, the selected variables obtained were, in increasing order: P_p(jl2), P_p(se3), P_p(no1), P_p(no3), T_n(ag2), T_x(ag2), Z(oc2), Z(no1), Z(no2), α(oc1). Consequently the following common variables were obtained from the two modes: α(oc1), Z(no2), Z(oc2), T_x(ag2) and P_p(jl2). Again, this result reinforces the importance of Z and α in predicting wheat yield. PDSI and SWA were not included as valid variables.

The condition of "up to six variables" was added to the model in order to simplify their future application in crop assessment equations, giving rise to a higher degree of statistical significance. Of all the different combinations used, the best result for the multiple-regression equation is expressed by the α(oc1), Z(no1), T_n(se3), SWA(oc1) and P_p(jl2) variables (model 4D), characterized by an SEE of 178.9 kg ha^{-1} and an R^2 coefficient of 0.91. When

Table 2 Selected variables for 10-day (D) and phenological (P) models, using meteorological (*met*), processed (*pro*), indices (*ind*), and total (*tot*) variables as in Table 1 and variables truncated (*trun*) in time, the coefficient of determination R^2, the standard error of estimate *SEE* (kg ha^{-1}) and the root-mean-square error *RMSE* (kg ha^{-1}). The models showing the best fit are indicated by *bold type*

Models	Variables						R^2	SEE	RMSE
1D(met)	T_n(jl2)	T_n(ag3)	T_x(oc2)	T_n(no3)	P_p(ag2)	P_p(no2)	**98.1**	**90.5**	**345.7**
2D(pro)	α(oc1)	α(oc2)	α(se2)	α(ag2)	SWA(oc1)	α(no1)	92.0	186.8	501.8
3D(ind)	Z(se3)	Z(no2)	PDSI(jn3)	Z(ag2)	Z(no3)	Z(ag1)	76.9	318.2	365.8
4D(tot)	α(oc1)	Z(no1)	T_n(se3)	SWA(oc1)	Pp(jl2)	–	**91.9**	**178.9**	**470.0**
5D(tru)	α(oc1)	P_p(ag3)	P_p(se3)	P_p(oc2)	α(oc2)	P_p(se1)	82.5	276.7	668.6
1P(met)	P_p(7)	T_x(7)	P_p(6)	P_p(1)	T_n(3)	–	87.2	224.2	791.3
2P(pro)	α(4)	α(7)	SWA(4)	SWA(6)	α(5)	–	84.1	250.2	740.0
3P(ind)	Z(7)	PDSI(1)	Z(3)	Z(1)	–	–	61.8	373.5	>1000
4P(tot)	α(4)	Z(7)	SWA(4)	SWA(6)	T_x(4)	–	**88.9**	**209.5**	**440.0**
5P(tru)	α(4)	T_x(4)	P_p(1)	SWA(4)	SWA(5)	–	**84.0**	**250.6**	**491.3**

verified against observed yields from 1995 to 1998, a RMSE of 470 kg ha^{-1} was obtained.

Truncated model

An equation with four variables: α(oc1), Z(oc3), α(ag1) and P_p(oc3) was used to obtain an "early assessment model" truncated at the end of October (model 5D), with SEE equal to 287 kg ha^{-1}, R^2 equal to 0.72 (F to-enter: 4.85 and minimum tolerance: 0.200). With the introduction of more severe conditions only two variables were included: T_x(oc1) and Z(oc2). In this case SEE increased to 410 kg ha^{-1}, despite an R^2 value of 0.76. In both cases, as expected for truncated models, the testing procedure gave RMSE values greater than 670 kg ha^{-1}.

Phenological phase models

For the phenological phase scale the same steps as for the 10-day time scale were followed. The results of the correlation analysis are summarized in Table 3. Statistical results establish that grain yields are strongly related to the amount of rainfall during emergence, stem elongation, and flowering. A positive correlation with T_n is observed during the vegetative period, which is in agreement with the phenological concept that low temperatures during this stage favor the rate of onset of primordial leaves. Low temperatures at flowering and during the following 10 days begin to affect the crop negatively. High temperatures during the stalk elongation and heading stages show a negative correlation with yield, confirming that high temperatures during these phases are detrimental to wheat growth. The influence of hydrological conditions on wheat can also be analyzed by means of the α coefficient. Mean values of $\alpha \geq 0.50$ characterize the study region, 0.5 being the minimum water requirement for establishment of a crop. Total water storage (SWA) is increasingly important from stalk elongation onwards. Soil water storage during the pre-sowing stage does not show a correlation with crop growth. The indices Z and PDSI were designed as meteorological indices to measure soil water conditions. By definition PDSI shows a high correlation with the previous moisture availability. Both variables are closely related to the flowering and grain-filling stages.

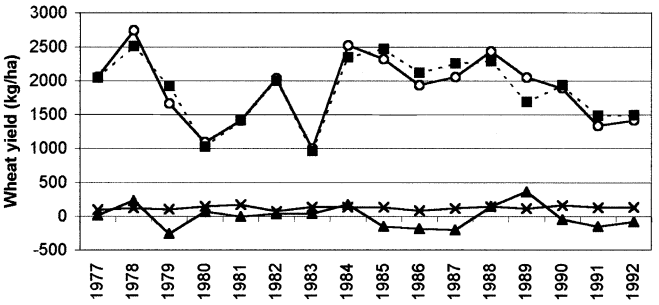

Fig. 5 The two upper lines are predicted (– – –, ■) and observed (–, ◯) grain yields. The lower lines represent residuals (×), and predicted standard error (▲) (kg ha^{-1}), for model 4P

The best selection of variables in terms of crop phases is (model 1P): $P_p(7)$, $T_x(7)$, $T_n(3)$, $P_p(6)$, $P_p(1)$, with R^2 = 0.87 and SEE = 224.2 kg ha^{-1}. Model 2P is composed of: $\alpha(4)$, $\alpha(7)$, SWA(4), SWA(6) and $\alpha(5)$, with a fitted regression where R^2 = 0.84 and SEE is 250.2 kg ha^{-1}. Finally, model 3P is formed with $Z(7)$, PDSI(1), $Z(3)$ and $Z(1)$ variables, with R^2 = 0.62 and SEE = 373.5 kg ha^{-1}.

All-variables phenological model

The best equation obtained to express wheat yield assessment, $\hat{Y}$, is:

$$\hat{Y} = 2628.0 + 2273.3\,\alpha(4) + 6.1\,Z(7) - 17.8\,\mathrm{SWA}(4) + 11.9\,\mathrm{SWA}(6) - 101.7\,T_x(4) \quad (2)$$

with R^2 = 0.88 and SEE = 209.5 kg ha^{-1} (model 4P).

Figure 5 shows the relation between predicted and observed values, residuals, and predicted SE for this model (4P) using phenological-scale and mixed parameters.

The analysis of residuals as a function of observed yields shows a tendency to under-estimate yields for low and medium crop-yield values. The control test assessment for this model was measured, showing RMSE to account for 440.0 kg ha^{-1}, which constitutes the best result among all the P models considered.

Table 3 Correlation coefficients between grain yields and agroclimatic variables as in Table 1, using phenological stages. See text for phase definitions. Enhanced values (*bold type*) indicate a significance level of 5%

Period	P_p	T_n	T_x	α	SWA	Z	PDSI
Pre-season mr1–jn2	–0.1210	**–0.4359**	–0.1658	0.0162	0.0488	–0.1559	0.0919
Planting jn3–jl1	0.0661	0.3584	0.1438	–0.1193	–0.1157	–0.0745	–0.1367
Emergence jl2–ag3	**0.3677**	0.2168	–0.2975	0.1627	0.1503	0.3119	0.0558
Tillering sp1–sp3	0.2737	0.2317	–0.2844	0.3526	0.2600	0.3429	0.1249
Stem elong. oc1–oc2	**0.5412**	–0.0371	**–0.5480**	**0.6990**	**0.4833**	**0.5870**	0.3702
Heading oc3	0.2528	–0.0742	–0.2554	**0.4484**	0.5081	0.3185	**0.4363**
Flowering no1	**0.4493**	–0.2292	–0.3083	0.3576	**0.5311**	0.4452	0.4743
Grain fill. no2–no3	0.0400	–0.0936	–0.2499	**0.5964**	0.5961	0.6125	0.5342
Maturity de1–de2	–0.2865	**–0.4058**	–0.0910	0.0650	0.1072	–0.2612	0.2162
Harvest de3	0.2570	–0.2490	–0.0156	–0.2280	0.2620	0.2249	0.2562

Truncated phenological model

A combination of several variables and phenological stages was considered in an early-yield assessment model. The various results pointed to the convenience of incorporating the parameters α, P_p, SWA and T_x. A model including $\alpha(4)$, $T_x(4)$, $P_p(6)$, SWA(4) and SWA(5) and using the first six phenological phases resulted in an R^2 equal to 0.84 (model 5P). Despite a higher SE (250.6 kg ha^{-1}) and RMSE (491.3 kg ha^{-1}) compared with model 4P, Model 5P is considered to be a good assessment of early yields. This truncated phenological model proved to be better than model 5D, on the 10-day scale.

Multivariate model results

Table 2 presents a summary of the best results for the multivariate regressions tested, showing the different models obtained for 10-day (D) and phenological (P) scales. Numbers 1–5 in Table 2 refer to the quality of the variables employed: meteorological, processed, indices, total or mixed variables and truncated. In all the D models except 5D, the determination coefficient proved to be higher and the standard error lower than in the phenological models. Among the latter, 4P shows the highest R^2 value and the lowest standard error of estimate. This indicates an a priori preference for D models. When testing the estimated yields, the RMSE value reflects the predictive accuracy of the models and demonstrates the good performance of models 1D and 4D for a D scale and 4P and 5P for phenological stages.

The multiple-regression equations for the models 1D, 4D, 4P and 5P are summarized in Table 4. A model's constant (or interception) is a measure of the mean yield value but, as in 1D and 4D, does not always represent real conditions. Furthermore, the multiple determination coefficient, R^2, is a measure of the extent to which X_i determines $\hat{Y}$ (Eqn. 1) and, while a useful diagnostic tool, it should not be used as the only one to assess the utility and goodness-of-fit of a model (von Storch and Zwiers 1999). There are several reasons for our selection of model 4P (Eqn. 2) as the best model: it has a reasonable constant value, the five variables are related to water availability and temperature range for recognized crop stages, it has adequate residuals and, according to the RMSE, is a good estimator of wheat yields. Although model 1D would appear to be the best option using a 10-day scale, it was discarded because it gives rise to an unrealistic interception constant and incidence of minimum temperature during the third period (November) that do not respond to a limiting condition for the region.

Table 4 Interception value (constant), variables (as in Table 1) and coefficients for the best multivariate regression models as mentioned in the text

Model	1D(met)	4D(tot)	4P(tot)	5P(trun)
Interception (kg ha^{-1})	9079.4	249.3	2843.5	2628.5
Variables	T_n(jl2)	Pp(jl2)	$\alpha(4)$	$\alpha(4)$
Coefficients	23.9	13.8	2532.3	2532.3
Variables	P_p(ag2)	T_n(se3)	SWA(4)	SWA(4)
Coefficients	21.24	108.1	–18.0	–17.9
Variables	T_n(ag3)	SWA(oc1)	$T_x(4)$	$T_x(4)$
Coefficients	–134.9	–8.1	–101.8	–130.4
Variables	T_n(oc2)	α(oc1)	SWA(6)	SWA(5)
Coefficients	–215.2	1714.6	11.9	10.0
Variables	P_p(no2)	Z(no1)	$Z(7)$	Pp(6)
Coefficients	5.6	8.1	6.1	11.0
Variables	T_n(no3)	–	–	–
Coefficients	–203.0	–	–	–
Adj R^2	0.9688	0.8781	0.8330	0.8090
R^2	0.9813	0.9187	0.8886	0.8727

Discussion

There are a number of possible statistically significant correlations between variables that fit the data well, but a valid predictability criterion must have variables that are applicable to the various stages of crop development.

Multiple-regression models are valuable tools as long as they are based on easy-to-obtain variables. The coefficient of determination is used to describe the ability of the fitted linear model to explain variations in yield data. Residual analysis helps to show systematic departures from the fitted line and detect the outliers. The first step in the different analyses performed was to choose variables and time scales shorter than 1 month that could subsequently be re-dimensioned and adapted to crop phases. We then went on to study the relations between dependent and independent variables. Variables were taken as independent even though the nature of meteorological variables is to a certain extent dependent: the processed variables and indices, for example, are related to water availability and through radiation to temperature. Finally, many trials were carried out in order to determine which variables were most useful and to select the models.

In this study variables were analyzed in a linear form, and no attempt was made to include crossed variables or, exponential or potential functions. When working with all the variables (models 4D and 4P), it is interesting that α and Z were the first two to be selected for the model, confirming the results of exploratory models (see Results: All-variables 10-day model). The use of a very elaborate variable like PDSI leads to an inadequate data fit; raw meteorological variables and those from a simple soil water balance, on the other hand, give better results (Table 2, 1D, 2D). Nevertheless, other authors have found that the mean monthly PDSI index has a good correlation with wheat yield (Akinremi and McGinn 1995). In our study there is no statistical justification for including preseason variables in the selected models. Model 1D, using meteorological variables, shows a good fit to the values but the coefficients are somewhat unrealistic.

In testing the predictive ability of the regression models, unsatisfactory results were obtained for those years in which meteorological extremes occurred. This can be explained by the fact that, in applying regression methods, outliers are generally discarded and extreme values are smoothed. High-risk meteorological extremes, such as late frosts (as in 1998) and heavy rains during physiological maturity (as in 1996), might be included in future models.

The application of complex simulation models such as CERES wheat to the pampa region proves that the differences between observed and predicted yields do not exceed the standard deviation (660 kg ha^{-1}) (Magrin et al. 1991). Simple regression models like some of those presented in the present study can therefore be of practical use in yield prediction and as a tool for assessing the impact of climatic change.

Conclusions

The present study shows that environmental variables on 10-day and phenological scales can be used to design multiple-regression crop models for wheat yields in Bordenave. The models could also be appropriate to estimate wheat yields for the neighboring region. The coefficient of determination (R^2) obtained by stepwise regression in the 10-day models is generally higher than that from models based on crop phases. The model with simple averages of temperature and precipitation totals (model 1D, Table 2) provides a more meaningful representation of crop conditions than models with indices as variables. For truncated models, a better predictive equation is obtained for phenological stages (model 5P, Table 2) than for 10-day intervals. However, when mixed parameters are used, the stepwise method generates good results for both models, 4D and 4P (Table 2). Model 4P (Eqn. 2), which includes mixed variables from stem elongation to grain filling, was finally selected as the optimum model. The selected variables relate to soil water availability and extreme temperatures, which contribute to the total variability of yield. According to the RMSE value obtained, the use of phenological stages improves yield assessment, particularly for those years with extreme meteorological conditions. One possible explanation for this is that phenological phases include several 10-day periods, moderating the effect of extreme events that have little probability of occurring and are of short duration. When extreme values are present for a 10-day interval not included in the model, predictions may fail. These models can be improved by incorporating terms related to the incidence of late frosts and intense rainfall events. In view of the residuals obtained, it might be useful for future experiments to introduce quadratic terms.

Finally, mention should be made of the limitations and restrictions of the present study. These relate to: (1) the reduced number of crop years, (2) the fact that the models can not been applied outside the region, and (3) the fixed starting and ending dates of the phenological phases.

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